

Absolute Error and MAE

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can calculate MAE MSE and RMSE.

I can explain how squaring changes sensitivity to large errors.

I can match the loss and units to the task.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

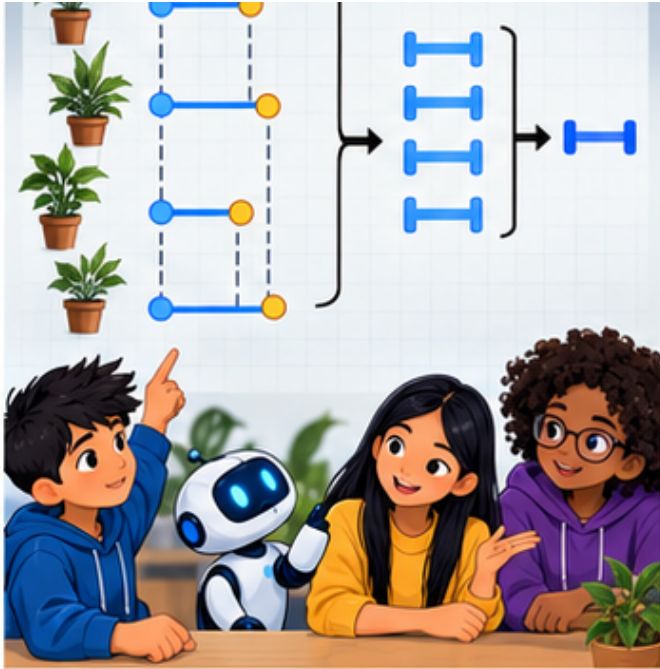
Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Absolute error measures distance from truth without cancellation; MAE averages those distances in the target's original unit.



LEARNING TARGET Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Builder name and experiment date:

Readiness for the connected explanation

Which constant minimizes training squared error?

This week's end goal

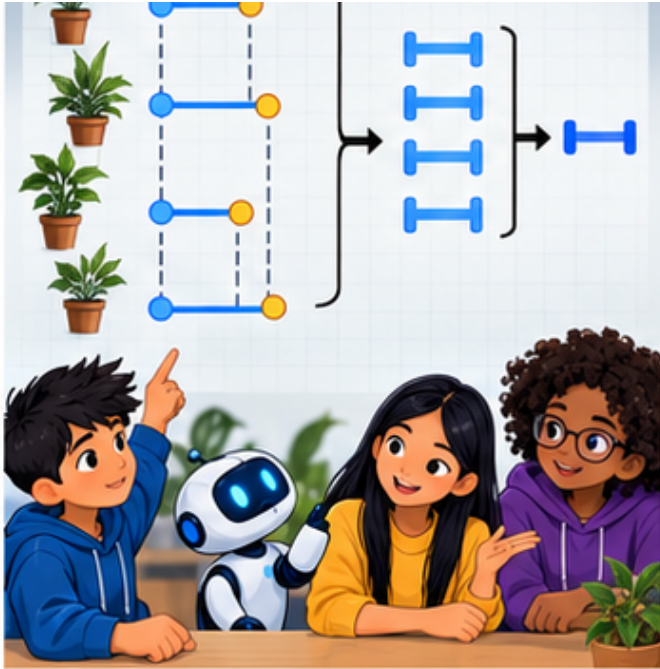
By the end, I can explain absolute error and mae using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use absolute error and mae to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A loss summarizes error magnitude without positive and negative cancellation. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly. RMSE takes the square root of MSE so it returns to the target unit. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines absolute and squared loss.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

IN

INPUT

Calculate residuals

ML

MODEL

Absolute error measures distance from truth without cancellation; MAE averages those distances in the target's original unit.



OUT

EVIDENCE

and mean absolute error.

QUESTION Calculate residuals, absolute errors, and mean absolute error.

MODEL IDEA Absolute error measures distance from truth without cancellation; MAE averages those distances in the target's original unit.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual heights: 10, 14, 20; predictions: 12, 13, 17. Residuals: -2, +1, +3 cm; absolute errors: 2, 1, 3 cm.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Residuals [-1, 2, -3] cm have absolute errors [1, 2, 3], so $MAE = 6/3 = 2$ cm. Squared errors [1, 4, 9] sum to 14, so $MSE = 14/3$ square cm and $RMSE = \sqrt{14/3}$, about 2.16 cm. Keep the square and the average in the correct order.

```
from math import sqrt
residuals = [-1, 2, -3]

mae = sum(abs(r) for r in residuals) / len(residuals)
mse = sum(r * r for r in residuals) / len(residuals)
print(mae, mse, sqrt(mse))

# About 2.0, 4.67, 2.16.
# MAE and RMSE: cm. MSE: square cm.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs

and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Connect the picture and the explanation

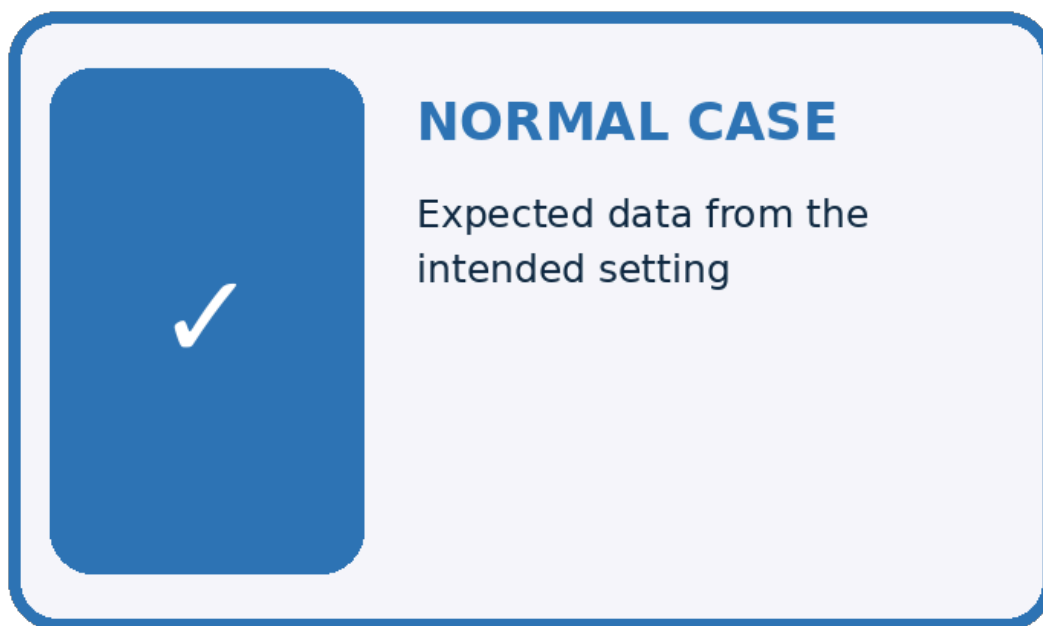
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





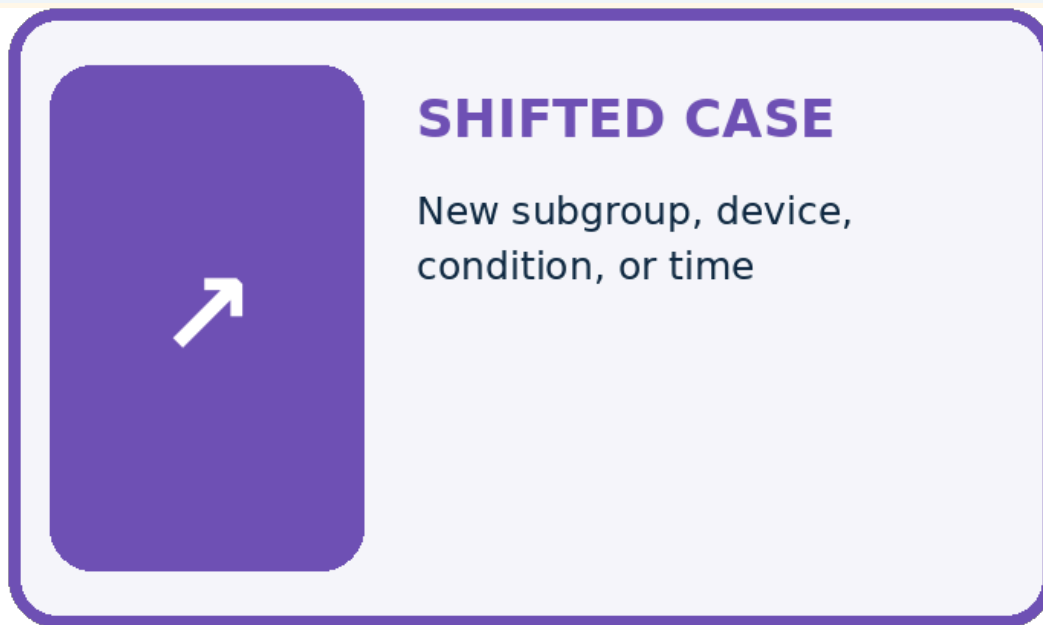
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Convert every residual to millimetres.

Prediction: MAE becomes 20 mm; MSE becomes $1400/3 \text{ mm}^2$; RMSE becomes about 21.60 mm.

Reason: Residuals scale by 10; squares scale by 100, while the square root returns a length unit.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

ABSOLUTE ERROR

absolute value of actual
minus predicted

2

MAE

mean of absolute errors

3

UNIT

measurement scale retained
by MAE

4

CANCELLATION

positive and negative errors
offset

Term · Precise meaning · My ML example

Term: absolute error

Precise meaning: absolute value of actual minus predicted

My ML example: _____

Term: MAE

Precise meaning: mean of absolute errors

My ML example: _____

Term: unit

Precise meaning: measurement scale retained by MAE

My ML example: _____

Term: cancellation

Precise meaning: positive and negative errors offset

My ML example: _____

MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average absolute error.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: MAE

Meaning: Mean absolute error

Word: MSE

Meaning: Mean squared error

Word: RMSE

Meaning: Square root of mean squared error

Word: Loss

Meaning: A numerical penalty for prediction errors

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 21 • WORKED EVIDENCE

- 1 Actual heights: 10, 14, 20; predictions: 12, 13, 17.
- 2 Residuals: -2, +1, +3 cm; absolute errors: 2, 1, 3 cm.
- 3 $MAE = (2 + 1 + 3) \div 3 = 2$ cm.
- 4 Compare with a baseline on the same cases and inspect the largest error.

STEP 1 Actual heights: 10, 14, 20; predictions: 12, 13, 17.

STEP 2 Residuals: -2, +1, +3 cm; absolute errors: 2, 1, 3 cm.

STEP 3 $MAE = (2 + 1 + 3) \div 3 = 2$ cm.

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Absolute and squared loss at a glance

Part • Evidence or action • Interpretation

Part: Residuals

Evidence or action: minus 1 plus 2 minus 3

Interpretation: Signed differences

Part: Absolute values

Evidence or action: 1 2 3

Interpretation: MAE 2 cm

Part: Squares

Evidence or action: 1 4 9

Interpretation: MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Give MSE and RMSE with units. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Absolute Error and MAE

Compare absolute and squared error without losing track of units.

The facts we will use

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Absolute errors

Absolute values 1, 2, 3 give $MAE 6/3 = 2$ cm.



2. Squared errors

Squares 1, 4, 9 give $MSE 14/3$ cm², approximately = 4.67 cm².



3. Root squared error

$RMSE = \sqrt{14/3}$, approximately 2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Absolute errors

What happens and why: Absolute values 1, 2, 3 give $MAE 6/3 = 2$ cm.

Step: 2. Squared errors

What happens and why: Squares 1, 4, 9 give $MSE 14/3$ cm², approximately = 4.67 cm².

Step: 3. Root squared error

What happens and why: $RMSE = \sqrt{14/3}$, approximately 2.16 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. $RMSE = \sqrt{14/3}$, approximately 2.16 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Different loss functions weight large misses differently; a smaller selected loss is not automatically better for every consequence.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Average signed residual is a safe replacement for average absolute error. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause / next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Average signed residual is a safe replacement for average absolute error.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Choose and document the comparison metric before reviewing final-test results. Neither MAE nor MSE reveals all subgroup patterns or proves safety. Do not remove a large valid error solely because it worsens squared loss.

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Changing cm to mm multiplies residuals and MAE/RMSE by 10, but MSE by 100. A loss decrease across reports is not meaningful unless units, evaluated rows and loss definitions match. Retain the per-record residuals for investigation.

Practise with less help: Compute squared loss

Use workbook W2 for the connected example “Absolute and squared loss”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Absolute and squared loss. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

MAE Lab: calculate each row, verify units and denominator, then compare model and baseline on identical examples.

1. Write the hypothesis and expected evidence before running.
2. MAE Lab: calculate each row, verify units and denominator, then compare model and baseline on identical examples.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

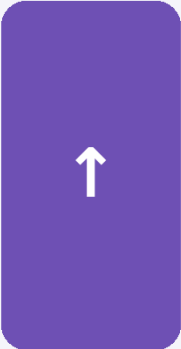
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: absolute error, MAE, unit, cancellation.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Average signed residual is a safe replacement for average absolute error.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

THIS WEEK'S CHECKPOINT Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Calculate residuals, absolute errors, and mean absolute error.

REQUIRED PROOF Calculate residuals, absolute errors, and mean absolute error.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Train a Regression Model.

13. Study a complete case

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Actual=[3,5,9], predicted=[2,7,6] cm.

The question

Calculate MAE.

Worked explanation

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Absolute errors are measured before averaging

Evidence or stage · Value or result

Evidence or stage: Case 1, actual 3, predicted 2

Value or result: 1

Evidence or stage: Case 2, actual 5, predicted 7

Value or result: 2

Evidence or stage: Case 3, actual 9, predicted 6

Value or result: 3

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

14. Try a new case independently

Independent case: Apply independently

Residuals [-2, 0, 4] m: calculate MAE MSE RMSE and the result of converting to cm.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Actual heights are 4, 8 and 10 cm; predictions are 5, 6 and 13 cm. Calculate absolute errors and MAE.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.